Part I: Theory Questions

(Chapter-wise Collection of All Theory Questions)

Module 1: Introduction to Engineering Graphics

1. **(W-24, S-20)** Give illustration and application of type of lines / List different types of lines used in engineering graphics with their applications.
2. **(W-24)** Explain reducing scale. Give two practical applications of it.
3. **(W-24)** Draw symbol of following and right use of it: 1) Cutting Plane line 2) Dimension line 3) Break line 4) Section Line.
4. **(S-25)** Explain the difference between 1^{st} angle and 3^{rd} angle projection method.
5. **(S-25, W-19, S-22)** Define Representative Fraction (R.F.). With reference to Representative Fraction, classify the scales / Explain plain scale and diagonal scale.
6. **(S-25, W-22, S-19)** Differentiate between aligned system and unidirectional system of dimensioning / Explain the systems of dimensioning.

Module 2: Engineering Curves

1. **(W-24)** Give Conic Terminology with sketch.
2. **(W-24)** Give classification of Engineering Curves.
3. **(W-24, S-20, S-22)** Define ellipse. What is eccentricity? What is the value of eccentricity for ellipse? / Define Conic sections. How are they classified based on eccentricity.
4. **(S-25, S-19)** Define the following curves: involute, ellipse and cycloid.
5. **(S-25)** Enlist methods for drawing Ellipse.
6. **(W-21)** What is an Involute? State its applications.

Module 3: Projections of Points and Lines

1. **(S-25)** Enlist types of lines used in projection.
2. **(W-19)** What do you mean by 'True Length' of a line? When does a line show its true length in HP and VP?

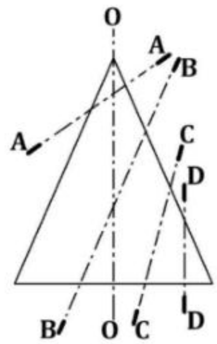
Module 5: Projections of Solids

1. **(W-24, W-19)** Draw the following sketches: Truncated Cylinder, Frustum of a cone and Frustum of a square pyramid.
2. **(W-24)** Define solid of revolution? Give three example of solid of revolution.
3. **(S-25, S-23)** Classify solids with shapes of drawings.
4. **(S-25, S-19)** Differentiate between: 1) Square pyramid and tetrahedron 2) Cube and square prism.
5. **(S-25, S-19)** Differentiate between prism and pyramid.
6. **(S-20, S-22)** Define: Polyhedron, Regular Polyhedron. List five regular polyhedrons / Define right solid, oblique solid and regular solid.

Module 6: Section of Solids

1. **(W-24)** Give different types of section planes.
2. **(W-24)** When we cut a cone using cutting planes as shown in figure, name the curves which are available as cut section of cone with particular cutting plane.

   [Refer to Figure from Paper: Q.2(b) Diagram, Winter 2024]



3. **(W-21)** What is a Section Plane? What is True Shape of Section?
4. **(W-18)** Define apparent shape and true shape with diagram.

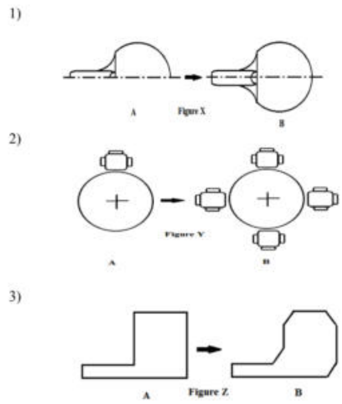
Module 8: Orthographic Projections

1. **(W-24)** Give Symbols for Methods of Projection.
2. **(W-24)** Give difference between two dimensional method with drawing.
3. **(S-25, S-20)** Write the comparison between first angle projection method and third angle projection method.
4. **(W-19)** Why second and fourth angle projection methods are not used?
5. **(S-20)** Write any three main differences between first angle and third angle projection system.

Module 10: Computer Aided Design (CAD)

1. **(W-24)** Which AUTOCAD command can be used to create Part B from Part A for the following figures? 1) Figure X 2) Figure Y 3) Figure Z.

   [Refer to Figure from Paper: Q.5(a) OR Diagram, Winter 2024]



2. **(W-24, W-22, S-20, W-23)** Advantages/Benefits of AUTOCAD over conventional drawing / State the importance of AutoCAD in engineering field.
3. **(S-20)** List any four editing commands in AutoCAD and explain their function.
4. **(W-21)** Explain the function of the following AutoCAD commands: MIRROR, ARRAY, TRIM, EXTEND.
5. **(W-24, W-18)** Explain use of following AUTOCAD commands: (1) MOVE (2) FILLET (3) OFFSET (4) CHAMFER.
6. **(W-18)** Why chamfer is done on work piece? Write the steps to create chamfer in AUTOCAD.
7. **(W-18)** List and explain different methods to draw circle in AUTOCAD.
8. **(W-18)** Write difference between line, polyline and its uses in AUTOCAD.
9. **(S-19)** Explain the following autocad commands: hatch, circle and array.
10. **(S-19)** Explain the following autocad commands: mirror, trim, extend and fillet.
11. **(W-19)** List and explain different methods to draw arc in AUTOCAD.
12. **(W-19)** List the six Essential Commands of Modify Panel in AutoCAD.
13. **(W-19)** List and explain different methods to draw Polygon in AUTOCAD.
14. **(S-22)** What are the various applications of Auto CAD?
15. **(S-23)** Explain the importance of file extension in the communication of drawing.
16. **(S-23)** Enlist any six CAD drawing preparatory commands with its symbols and function.
17. **(S-23)** Identify the command, that used 1) to remove certain unwanted part, 2) to show sectional view, 3) to make a continuous line and 4) to make a circular shape at the edge of drawing.
18. **(W-23)** Explain use of following AUTOCAD commands: 1. TRIM 2. ARC 3. MIRROR 4. GRID

Part II: Drawing/Practical Problems

(Chapter-wise Collection of All Drawing Problems)

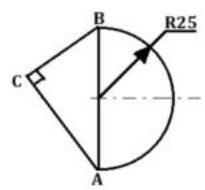
Module 1: Introduction to Engineering Graphics (Drawing)

1. **(W-24, S-20)** Construct a diagonal scale of $R.F. = 1/36$ (or similar) showing yard, foot and inch. Scale should be long enough to measure 5 yards. Indicate on it 3 yards 2 feet 9 inches.
2. **(W-24, W-22)** Construct a scale 1:50 to read up to 6 metres in meters and decimetres. Show on it a distance of 4.9 metres.
3. **(W-24)** The distance between two trees on road is 876m. Show it on a scale of 1:6250. Also show distance of 653m distance between two vehicles parked on a side of the road.
4. **(S-25, S-24)** Draw regular pentagon and hexagon of 30 mm side using UNIVERSAL METHOD.
5. **(S-25, S-24, S-22, W-22)** The distance between two towns is 250 km and is represented by a line of length 50mm on a map. Construct a scale to read 600 km and indicate a distance of 530 km on it.
6. **(S-20)** Construct a diagonal scale of R.F. = 1/4000 to show meters and long enough to measure up to 500 meters. Mark a distance of 374 meters on it.
7. **(W-18)** For 100 cm of a line compare size of drawing length on basis of full scale, reducing scale & enlarged scale.
8. **(S-19)** Draw a diagonal scale of R.F, 1:5 showing decimeters, centimeters and millimeters and long enough to measure upto 8 decimeters. Show a distance of 5.35 dm.
9. **(W-19)** Construct a scale of $1cm = 1$ m to read meters and decimeters and long enough to measure up to 14 meters. Show on this scale a distance equal to 12.4 meters.
10. **(W-23)** Construct a scale of 1:40 to read meters and decimeters and long enough to measure 6 m. Mark on it a distance of 4.7 m and 3.2 m.

Module 2: Engineering Curves (Drawing)

1. **(W-24, S-19)** The major and minor axes of an ellipse measure 100 mm and 70 mm respectively. Draw an ellipse by the rectangle method.
2. **(W-24)** Draw involute of a string unwound for the given figure from point C, where $AC = 40$ mm and $BC = 30$ mm.

   [Refer to Figure from Paper: Q.2(c) Diagram, Winter 2024]



3. **(W-24, S-22)** Draw an ellipse using concentric circle method. Given the distance between Foci is 90mm and length of major axis is 130mm.
4. **(S-25)** Draw an Archemedian spiral of 1.5 convolutions, the largest & the smallest radius being 55 mm & 10 mm respectively. Draw tangent & normal to the spiral at a point 40 mm from the pole.
5. **(S-25, S-22, W-19)** The foci of an ellipse are 120mm apart. The minor axis 72 mm long. Determine the length of the major axis and draw ellipse by oblong method.
6. **(W-19)** A circle of 50 mm diameter rolls along a straight line without slipping. Draw the curve traced out by a point P on the circumference, for one complete revolution of the circle. Name the curve.
7. **(S-20, S-19)** Draw an epicycloid generated by a rolling circle of diameter 40 mm and the radius of the directing circle is 120 mm. Draw tangent and normal to the curve at any point on it.
8. **(W-18)** A point P moves towards another point O, 90 mm from it, and reaches it during 1.5 revolutions around it in clockwise direction. Its movement towards O is uniform with its movement around it. Draw the curve traced out by the point P and name it.
9. **(W-18)** A fixed point is 54 mm away from a fixed straight line. Draw the locus of a point P moving in such a way that the ratio of its distance from the fixed straight line is 5:4. Name the curve.
10. **(S-19)** In the mechanism shown in Fig.1, the connecting rod is constrained to pass through the trunnion at D. Trace the locus of the end C and a point P on BC for one complete revolution of the crank. In Fig. 1 consider AB as 30 mm.

    [Refer to Fig. 1 from Paper: Summer 2019]

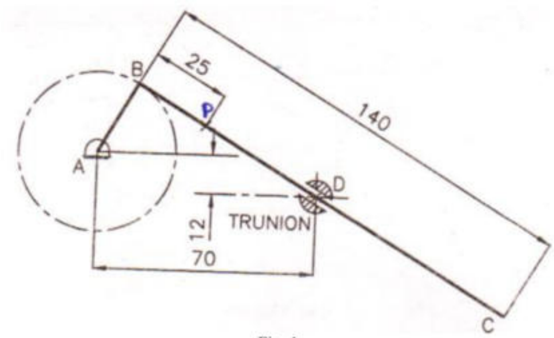


Fig. 1

11. **(S-19)** A triangle ABC has sides $AB = 75$ mm, $BC = 60$ mm and $CA = 75$ mm. Draw a parabola passing through points A, B and C when side BC is horizontal.
12. **(W-19)** OAB is a slider crank mechanism. Slider B is sliding on a straight path passing through O as shown in Figure 1 given below. Crank OA is 30 mm and rotates in anticlockwise direction and length of connecting red AB is 100 mm. A rod NP of 30 mm length is attached to AB such that $AN = 40$ mm and NP is perpendicular to AB as shown Draw locus of point P for one complete revolution of the crank.

    [Refer to Figure 1 from Paper: Winter 2019]

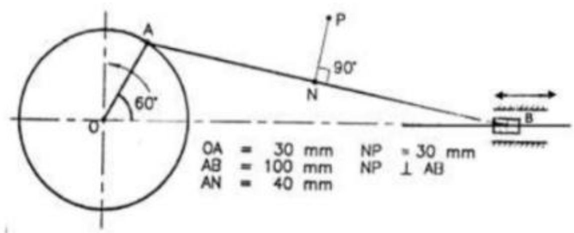


Figure 1.

13. **(W-19, W-23)** Draw a parabola by tangent method, given its base as 90 mm and the height of axis is 75 mm.

14. **(S-22)** Find the locus of a point P, moving in a plane, keeping its distances equal from two non-parallel fixed straight lines O_1O_2 and O_1O_3.

15. **(S-22)** Crank OA rotates about 'O'. (Slide along 'OC', rod BD is fixed to connecting rod AC). Draw locus of point 'D' for one complete rotation of crank OA.

16. **(S-22)** A circle of 50 mm diameter rolls on the circumference of another circle of 150 mm diameter and outside it. Draw the locus of the point P on the circumference of the rolling circle for one complete revolution of it. Take initial position... at the contact point between two circles

17. **(S-22)** A point P is 120 mm away from the fixed point pole O. A point P moves towards pole O and reaches the position P' in one convolution where OP' is 22 mm. The point P moves in such a way that its movement towards fixed point O, being uniform with its movement around fixed point pole O. Draw the curve traced out by the point P. Name the curve.

18. **(S-22)** Construct a parabola, when the distance of the focus from the directrix is 40 mm. Also draw a tangent and normal at a point on it 30 mm from F.

19. **(W-22)** In a slider crank mechanism OBA, the crank OB is 200 mm long and the connecting rod BA is 850 mm long. Plot the locus of point P where P is midpoint of connecting rod.

20. **(W-22)** Construct one complete turn of an involute of a circle having diameter 30 mm. Also draw tangent and normal to the curve S. The point S is a point on the involute and it is at a distance of 45mm from centre of the circle.

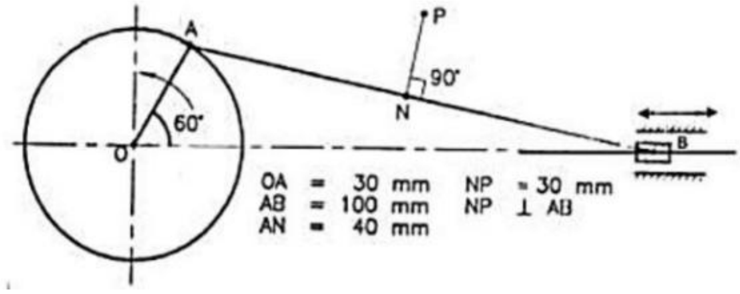
21. **(S-23)** Construct the Involute of hexagon of 25 mm sides.

22. **(S-24)** A pole is of a shape of half hexagon and semicircle. A string is to be wound having length equal to the pole perimeter Draw path of free end P of string when wound completely. (Take hexagon 30 mm sides and semicircle of 60 mm diameter.)

23. **(S-24)** Rod AB, 100 mm long, revolves in clockwise direction for one revolution. Meanwhile point P, initially on A starts moving towards B and reaches B. Draw locus of point P.

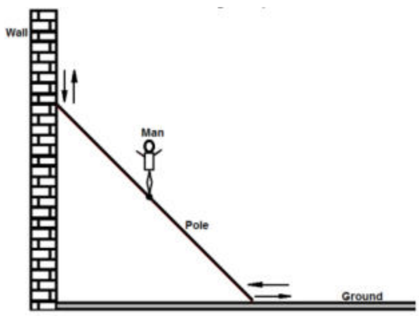
24. **(S-24)** Draw loci of point 'P' given in figure.

    [Refer to Figure from Paper: Q.2(c) OR Diagram, Summer 2024]



25. **(W-23)** A pole is sliding vertical to horizontal position as shown in figure. A man is standing at the midpoint of pole. Trace locus of man for sliding of the pole from vertical to horizontal. Length of pole is 480 centimeters.

    [Refer to Figure from Paper: Q.1(c) Diagram, Winter 2023]



26. **(W-23)** A thin circular disc of 50 mm diameter is allowed to roll without slipping from upper edge of sloping board which is inclined at 15° with the horizontal plane. Draw the curve traced by the point on the circumference of the disc. Consider the point is at contact surface of disc and sloping board initially.

27. **(W-23)** Major axis of an ellipse is 120 mm and the distance between focii of ellipse is 80 mm. Draw an ellipse using oblong method. Draw tangent and normal to any point to the curve.

Module 3: Projections of Points and Lines (Drawing)

1. **(W-24)** A line PQ 70 mm long has its end P in VP and end Q in HP. Line is inclined to HP by 60° and VP by 30^0 Draw the projections.

2. **(W-24)** A line AB has a point P on it such that AP: $PB = 1 : 2$ The end A is in the first quadrant and it is 20 mm above H.P. while the end B is in the V.P. The point P is 35 mm from the H.P. The line is inclined at 30° to the H.P. and the elevation length of the line is 70 mm. Draw the projections of the line AB and the point P. Find the true length, the plan length and the inclination of the line with V.P.

3. **(W-24)** Draw following projection of points: (1) Point 'A' is 20 mm above HP and 25 mm In front of VP. (2) Point 'B' is 35 mm behind VP and 30 mm above HP. (3) Point 'C' is in the HP and 15 mm In front of VP.

4. **(W-24, S-19)** Two lemons on a tree, planted near the compound wall of a bunglow, are 1.0 m and 1.25 m above the ground and 0.5 m and 0.75 m from a 15 cm thick compound wall but on the opposite sides of it. The distance between lemons measured along the ground and parallel to the wall is 1.0 m. Determine the real distance between centre of two lemons.

5. **(W-24, S-20)** The top view of a 75mm long line AB measures 65mm, while its front view measures 50mm. It's one end A is in HP and 12mm in front of VP. Draw the projections of AB and determine its inclination with HP and VP.

6. **(W-24)** Draw the projection of following points: 1. Point A is 20 mm above HP and 35 mm behind VP. 2. Point B is 20 mm below HP and 35 mm in front of VP. 3. Point C is in VP and 15 mm above HP.

7. **(S-25, S-24)** Draw following projection of points: (1) Point 'A' is 25 mm above HP and 20 mm Infront of VP. (2) Point 'B' is 35 mm behind VP and 40 mm above HP. (3) Point 'C' is in the HP and 10 mm Infront of VP.

8. **(S-25, S-24)** Draw projections of line $AB = 85$ mm inclined to 60° HP & 30ř VP.

9. **(S-25, S-19)** The distance between the end projectors of a straight-line PQ is 130mm. The end P is 40mm below HP and 25mm in front of the VP. Q is 75 mm above HP and 30 mm behind VP. Draw its projections. Find TL of the line.

10. **(S-25, S-24)** The elevation of line AB, 80 mm long, measures 55 mm. The end A is 20 mm above H.P. and 10 mm in front of V.P. Draw the projection of the line and finds its true inclination with H.P. and V.P, if the end B is 25 mm below H.P. and is behind V.P.

11. **(S-25, W-22)** Draw the projection of following points on the same X-Y line. (1) Point A is 25 mm above HP and 30 mm behind V.P. (2) Point B is 20 mm above HP and on V.P. (3) Point C is 10 mm bellow HP and 25 mm behind V.P. (4) Point D is on HP and 40 mm in front of V.P.

12. **(S-25)** A Line AB, 90 mm long, is inclined to H.P. by 30° and inclined to V.P. by 45° The line is in first quadrant with Point A 15 mm above H.P. and 25 mm in front of V.P. Draw the projection of line AB.

13. **(W-21)** A line AB, 75 mm long, has its end A 20 mm above HP and 25 mm in front of VP. The line is inclined at 45° to HP and 30° to VP. Draw its projections.

14. **(W-18)** Point P of a straight line PQ is 25mm above H.P. and point Q is 65 mm in-front of V.P. The line makes an angle of 30° with H.P. and its plan is 45° to the XY line. Draw the projections of the line if the plan length is 70mm. Also find the true length of the line and the angle made by the line with V.P.

15. **(W-18)** Draw projection of following points (i) Point R is 10 mm behind V.P. & 20 mm above H.P. (ii) Point S is in H.P. & 22 mm in front of V.P. (iii) Point T is 15 mm in front of V.P & 25 mm below H.P.

16. **(W-18)** A line AB, 75mm long, is parallel to VP and inclined to the HP, by an angle 45°. Point A is 30mm below HP and 20mm in front of VP. Point B is in the first quadrant. Draw the projections of the straight line AB.

17. **(W-18)** A line PQ, 100 mm long, is inclined at 30° to the HP and 45° to the VP. Its mid-point M is in the VP and 20mm above the HP. Draw its projections, when its end P is in the first quadrant and Q is in the third quadrant.

18. **(W-18)** Draw Projections of the following lines. (i) Line MN 50mm is in 1 quadrant and parallels both H.P. & V.P. (ii) Line PQ 35mm is in 3^{rd} quadrant and remains perpendicular to V.P. and parallel to H.P.

19. **(W-18)** The distance between the end projectors of a straight line PQ is 60mm. The line makes 30° and 45° angles with HP and VP, respectively. The end P is 30mm below HP and 50mm in front of the VP. Draw its projections when end Q is in third quadrant. Find TL of the line.

20. **(S-19)** A line AB is 100 mm long. It is inclined at 40° to the HP and 30^c to the VP. The end A is 10 mm above HP and 25 mm in front of VP. Assuming the end B in the first quadrant, draw the projections of the line AB.

21. **(S-19)** Draw the projections of the following points on the same x-y line: (i) Point A is 20 mm above the HP and 20 mm behind the VP. (ii) Point B is 40 mm above HP and 10 in front of VP. (iii) Point C is 25 mm below HP and 40 mm behind VP.

22. **(W-19)** Point A is 20 mm above HP and 30 mm in front of VP and point B is in the HP and 40 mm behind the VP. The distance between in their projectors is 50 mm. Draw the projections of the points. Also draw straight lines joining their top and front views.

23. **(W-19)** The end P of a line PQ 120 mm long is 30 mm above HP and 60 mm behind VP. The line is incline at angle of 30° with the reference plans of the projection. The point Q is below HP and behind VP. Draw the projections of line PQ and locate the point Q.

24. **(W-19)** Plan and elevation of a line AB, 80 mm long, measure 60 mm and 72 mm respectively. End A of the line is in HP and end B is in VP. Draw its projections, assuming the line to lie in first quadrant.

25. **(S-20)** Draw the projections of the following points on the same X-Y line. (a) Point 'A' is 20 mm below H.P and 20 mm in front of VP. (b) Point 'B' is 30 mm above H.P and 40 mm in front of VP. (c) Point 'C' is on VP and 30 mm above HP.

26. **(S-20)** A line PQ 85 mm long has its end 'P' 10 mm above HP and 15 mm in front of VP. The top view and front view of line PQ are 75 mm and 80 mm respectively. Draw its projections. Also determine the true and apparent inclinations of the line.

27. **(W-21)** Draw the projections of the following points on the same X-Y lime. (1) Point A 25 mm below H.P. and 20 mm in front of V.P. (2) Point B 35 mm above H.P. and 40 mm in front of V.P. (3) Point C on V.P. and 30 mm above H.P.

28. **(W-21)** Draw projection of line CD is 50 mm long, when it is not contained by any plane and parallel to both planes. A line CD is 20 mm above H.P. and 25 mm in front of V.P.
29. **(W-21)** A straight line AB 110 mm long is inclined at 30^0 to H.P. and 45° to V.P. One of its end A is 40 mm from the V.P. and 20 mm above H.P. Draw the projection of line AB.
30. **(W-21)** A line PQ 130 mm long is parallel to the V.P. and inclined at 60^0 to H.P. The end nearest to the H.P. is 30 mm from it and 40 mm from the V.P. Draw the projection of line PQ.
31. **(S-22)** A line PQ 100 mm long is inclined at an angle of 45° to H.P. and 30° to V.P. One of its end point $^*P'$ is in H.P. as well as VP. Determine its apparent inclination with V.P.
32. **(S-22)** A line PQ 70 mm long is parallel to V.P. and 30° inclined to H.P. The end P is 30 mm above H.P. and 20 mm in front of V.P. Draw the projections.
33. **(S-22)** A line AB, 75 mm long is inclined at an angle 35° to the H.P. and 55° to the V.P. Its end point 'A' is on the H.P. and 15 mm in front of the V.P. Draw the projections.
34. **(W-22)** A line MN, 60 mm long, is in V.P. It makes an angle of 30° with the H.P. The end point M is 15 mm above H.P. Draw projections of line MN. Also measure Plan length of the line.
35. **(W-22)** The line AB has its end A, 15 mm above H.P. and 10 mm in front of V.P. The end B is 60 mm above H.P. The distance between the end projectors is 50 mm. The plan of the line is inclined at 25° to X-Y line. Draw the projections of line. Find true length and inclination of the line with H.P.
36. **(W-22)** Draw the projections of a straight line 80 mm long inclined at 60ř to H.P. and 30° to V.P. with end A in the H.P. and the end B in the V.P.
37. **(S-23)** Draw the projections of the following points on the same ground lines, keeping the projectors 15 mm apart: (1) A in the H.P. and 20 mm behind V.P. (2) B 25 mm below the H.P. and...
38. **(S-23)** A line CD, 80 mm long is inclined at 45 to H.P. and 30ř to the V.P., its end C is in H.P. and 40 mm in front of V.P. Draw the projections.
39. **(S-23)** A 100 mm long line PQ is inclined at 30ř to H.P. and 45 to the V.P. Its mid-point is 35 mm above the H.P. and 50 mm in front of V.P. Draw its projections.
40. **(S-24)** Line AB 75mm long makes 45^0 inclinations with VP while it's FV makes 55°. End A is 10 mm above HP and 15 mm in front of VP. If line is in 1^{st} quadrant draw its projections and find it's inclination with HP.
41. **(S-24)** A room is of size 6.5m L,5m D,3.5m high. An electric bulb hangs 1m below the center of ceiling. A switch is placed in one of the corners of the room, 1.5m above the flooring. Draw the projections an determine real distance between the bulb and switch.
42. **(W-23)** A line PQ 70 mm long is parallel to profile plane and inclined at 30° to HP. Point P is 10mm Above HP and point Q is in VP. Draw its projections.
43. **(W-23)** A line AB, 80mm long is inclined to HP at 30° and to VP at 45°. The end A is in HP and VP. Draw projection of line AB. Find length of top view and front view of line and inclination of them with HP and VP.
44. **(W-23)** Draw projection of following points: 1. Point A is in HP and in VP 2. Point B is in VP and 25 mm below HP 3. Point C is 10 mm below HP and 20 mm in front of VP 4. Point D is 20 mm above HP and 20 mm in front of VP

Module 4: Projections of Planes (Drawing)

1. **(W-24, W-23)** A $30^\circ - 60^\circ$ set square has its shortest side 50 mm long and is in the H.P. The Top view of the set square is isosceles triangle. Draw the projections of the set square find its inclination with the H.P. / A 30ř-60ř set square of 60 mm longest side is so kept such that the longest side is in HP making an angle of 30° with VP. The set square itself is inclined at 45° to HP. Draw the projections of the set square.

2. **(W-24)** An isosceles triangle ABC having its base $AB = 40$ mm and altitude 60 mm is resting on the H.P. on Its base AB. Draw the projections of the plane when its surface is inclined to H.P. at an angle of 45^0 and the base AB which is on the H.P. is making an angle of 50° to the V.P.

3. **(W-24, S-22)** A rectangular plate PQRS. $25mm \times 40$ mm size, is in space with shorter edge parallel to H.P. and 15 mm above it. Plate PQRS is perpendicular to V.P. and inclined to H.P. by such an angle so that its plan becomes square. Draw the projections and find out its inclination with H.P. / A rectangular plane ABCD having 60 mm X 30 mm size is parallel to V.P., and perpendicular to H.P., and P.P. Draw the projections of the rectangle when it is 40 mm in front of V.P., and one of the smaller sides is parallel to H.P. and 20 mm above it.

4. **(W-24, S-24, W-23)** ABCD is a rhombus of diagonals AC 110 mm and BD 70 mm. Its corner A is in the H.P. and the plane is inclined to the H.P. such that the plan appears to be a square. Draw the projections of the plane and find its inclination with H.P. / A rhombus of 30mm and 50mm diagonal is resting on HP on one of its corner such that it appears a square of 30mm diagonals. Find its inclination with HP.

5. **(W-24, W-18, W-23)** A semi-circular thin plate, of 60 mm diameter, rests on the H. P. on its diameter, the surface is inclined at 30° to the H. P. Draw the projections of the plate.

6. **(W-24, W-23)** A pentagonal plane having edges 25 mm is placed on one of its corners on HP such that the surface makes an angle 30° with HP and side opposite to that corner on which the plane rests inclined at 60° to VP. Draw the top and front views of the plane.

7. **(S-25, S-19)** ABC is a triangle of sides $AB = 75$ mm, $BC = 60$ mm and $CA = 45$ mm. Its longest side is in H.P. Its surface makes an angle of 45° with the H.P. Draw its projections. / An isosceles triangular plate of 50 mm base and 75 mm altitude, appears as an equilateral triangle of 50 mm in top view. Draw the projections of a plate if its 50 mm long edge is on the HP and inclined at 45° to the VP. What is the inclination of the plate with the HP ?

8. **(S-25, S-24, W-22)** A pentagonal plate, side 25 mm is resting on H.P. on one of its corners with opposite edge to the corner making 30° with V.P. The plate is inclined to H.P. by 45°. Draw the projections of regular pentagonal plate. / A regular pentagon of 30 mm sides is resting on HP on one of its sides with its surface 45^0 inclined to HP. Draw its projections when the side in HP makes 30^0 angles with VP.

9. **(S-25, S-23, W-19)** A hexagonal plane of 30 mm side, rests on the V.P. on an edge such that the surface is inclined at 45ř to the V.P. Draw its projections. / A regular hexagon lamina ABCDEF has a side AB in VP and its side DE 45 mm in front of the VP and inclined to HP at 30°. Draw its projections.

10. **(S-25, W-22, S-23, W-21)** A circular plane having the diameter 50 mm is resting with point A of its periphery on H.P. The surface of the plane is inclined to H.P. such that the plan

of the plane becomes an ellipse with minor axis 30mm.Draw the projection of the plane when the plan of the diameter through point A is inclined at 30 to V.P.and the centre of the plane is 40mm from V.P.Find the inclination of the plane with H.P. / A circular disk, 50 mm in diameter, is resting on the H.P. on one of the point A of the circumference. Plane is inclined to the H.P. such a way that the top view of it is an ellipse of minor axis 40 mm. Draw its projections. / A circular plane of 80 mm diameter has one of the ends of the diameter in the H.P. while the other end is in the V.P. The plane is inclined at 30ř to the H.P. and 60ř to the V.P. Draw its projections.

11. **(S-20, W-18)** A square plate of 40 mm side rests on HP such that one of the diagonals is inclined at 30° to HP and 45° to VP. Draw its projections. / A square plate PQRS, edge 25mm, is in space with one of its corners in VP. Surface of the plate makes 50° with VP and it is perpendicular to HP. Draw its projections.

12. **(S-20)** A regular pentagon laminate of 30 mm each side is resting on HP on one of its sides with its surface making 45° with HP. Draw its projection when the side in HP makes an angle 30° with VP.

13. **(W-21)** A rectangular plane ABCD having 60 $mm \times 30$ mm size is parallel to H.P. and perpendicular to V.P. and P.P. Draw the projections of the rectangular when it is 40 mm above H.P. and one of the longer sides is parallel to V.P. and 20 mm in front of it.

14. **(W-21)** A square plane ABCD is of 30 mm side. It is kept on V.P. on one of its corner and it is inclined to V.P. at an angle of 30°. The surface of the plane is perpendicular to H.P. Draw the projection of plane.

15. **(W-22)** A hexagonal plate, side 25 mm, is inclined to H.P. by 45°. It is resting on HP on one of the corner. The plate is perpendicular to V.P. Draw projections of the plate.

16. **(W-22)** A square plate, 40 mm side, is resting on the H.P. on one of its sides. The plate is inclined to H.P such a way that it's top view looks like a rectangle with 20 mm shorter side. Draw its projections.

17. **(S-23)** The top view of a lamina whose surface is perpendicular to V.P. and inclined at an angle of 45ř to H.P. appears as a regular hexagon of 30 mm side, having a side parallel to the reference line. Draw the projections of plane and obtain its true shape.

18. **(W-23)** A square ABCD of 50 mm side is resting on HP on one of its side such that it appears as a rectangle of 35 mm X 50 mm. Draw its projection and find its inclination with HP.

Module 5: Projections of Solids (Drawing)

1. **(W-24)** A cylinder, diameter of base 60 mm and height 70 mm, is having a point of its periphery of base in V.P. with axis of cylinder inclined to V.P. by 45° and parallel to H.P. or ground. Draw the projections of the cylinder.

2. **(W-24, W-18, S-19)** A cone, diameter of base 60 mm and height 70 mm, has one of its generators in H.P. and making an angle of 45° with the V.P. Draw the projections of the cone when the apex is towards the observer.

3. **(W-24)** Draw the top view and front view of a rectangular pyramid of sides of base $40mm \times 50$ mm and height 70 mm resting with its base on H.P. such that one of the edges of the base is perpendicular to V.P.

4. **(W-24, W-23)** A square pyramid having 30mm edge of the base and axis of 70mm long and axis parallel to VP and inclined at 60^0 to HP. Draw projection of pyramid if one of the edge of its base inclined at 30° to VP and apex is on HP and 40 mm away from VP.

5. **(S-25, S-22)** A square pyramid, side of base 35 mm and axis length 50 mm is lying on the H.P. on one of its triangular faces. Draw the projections of the pyramid when the base edge contained by the triangular face on H.P. makes an angle of 45° on the V.P. keeping apex of the pyramid towards the observer.

6. **(S-25, S-24)** A cone 40 mm diameter and 50 mm axis is resting on one generator on HP. Draw it's projections.

7. **(W-22)** A pentagonal prism is resting on one of its corners of the base on HP. The longer edge containing that corner is inclined at 45° to HP. The axis of the prism is inclined at 30° to VP. Draw the projections.

8. **(W-19)** A right regular pentagonal pyramid, edges of base 25 mm and height 50 mm, has its base parallel to VP with one of its base edges in HP. Draw its projections. (FV and TV)

9. **(S-20)** A hexagonal prism with side of base 30 mm and axis length 60 mm is resting on one of its base edge on HP such that its axis is inclined at 45° with HP and the side on which it is resting is inclined at 30° with VP. Draw the projections.

10. **(W-21)** A square prism edge of base 35 mm and axis length 50 mm is resting on its base on the H.P. with an edge of base parallel to V.P. Draw the projection of prism.

11. **(S-23)** A square pyramid, side of base 40 mm and axis 60 mm is resting on its base on H.P. Draw its projections when (a) a side of the base is parallel to V.P., (b) a side of the base is inclined at 30ř to V.P., and (c) all the sides of the base are equally inclined to V.P.

12. **(S-24)** A rectangle ABCD 60 mm x 40 mm, is parallel to HP with one of its sides inclined at 30° to VP and the end of the side near to VP is 15 mm in front of the VP and 30 mm above HP. Draw its projections.

13. **(S-24)** A square pyramid, side of base 30 mm and height 50 mm, is resting on HP with a side of its base. Draw the projections if the axis is inclined to 45° to HP and parallel to VP

Module 6: Section of Solids (Drawing)

1. **(W-24, W-23)** A pentagonal pyramid, side of base 40 mm and height 80 mm, is resting on H.P. on its base with one of the edges of the base away from V.P. is parallel to VP. It is cut by an A.I.P. bisecting the axis, the distance of the section plane from the apex being 20 mm. Draw the elevation and sectional plan of the pyramid and draw the true shape of the section. Find the inclination of the section plane with the H.P.

2. **(S-25, S-23)** A hexagonal prism, edge of base 20 mm and axis 50 mm long, rests with its base on HP such that one of its rectangular faces is parallel to VP. It is cut by a plane perpendicular to VP, inclined at 45° to HP and passing through the right corner of the top face of the prism. (i) Draw the sectional top view.

3. **(S-20)** A cone of base diameter 50 mm and axis height 60 mm is resting on its base on HP. It is cut by a section plane perpendicular to VP and inclined at 45° to HP, passing through a point on the axis 20 mm from the apex. Draw the sectional top view and true shape of the section.

4. **(S-19)** A cylinder is resting on HP on its base. It is cut by AVP perpendicular to HP and inclined to VP by 45° and cutting it remaining 12 mm away from the axis. Draw the projections with section and draw also the true shape of the section. Take diameter of cylinder 55 mm and height 60 mm.

5. **(S-19)** A cone of 70 mm diameter of the base circle and 60 mm length of axis is resting on its base on the HP. It is cut by an AIP so that true shape of the section is an isosceles triangle with the vertex angle of 50° Set the required cutting plane and find its inclination with the HP. Draw sectional top view, front view and project the true shape of the section.

6. **(W-19)** One of the rectangular faces of the vertical square prism, 40 mm side and axis 65 mm long, make an angle of 30° to VP. It is cut by an AIP passing through the center of the axis in making an angle of 40° to HP. Draw front view, sectional top view and true shape of the section.

7. **(W-19, S-22)** A square pyramid, side of base 40mm and axis 60 mm long, has its base in HP and all edges of the base are equally inclined to VP. It is cut by a section plane perpendicular to the VP and inclined 45° to the HP such that it bisect the axis. Draw its sectional top view and sectional left side view.

8. **(S-20)** A square base pyramid with 40 mm side and axis 65 mm long, has its base on the HP and all the edges of the base are equally inclined to the VP. It is cut by a section plane perpendicular to the VP and inclined at 45° to the HP. Further, it is bisecting the axis of the pyramid. Draw its sectional top view, sectional side view and true shape of the section.

9. **(W-21)** A square prism side of base 30 mm and its axis is 50 mm is kept on H.P. on its base such that two side of base are perpendicular to V.P. It is cut by the horizontal cutting plane which cuts the axis at its midpoint. Draw front view and sectional top view.

10. **(W-22)** A cone, base diameter 30 mm and height 60 mm, is resting on its base. It is cut by AIP inclined at 45řto H.P. and passing through midpoint of the axis of cone. Draw front view, sectional top view and true shape of this sectional solid.

11. **(W-22)** A square prism, side of base 30 mm and height 45 mm, is resting on H.P. on one of the corners of the base. The longer edge containing that corner on which it rests, is inclined at 45° to H.P. and top view of it is inclined at 30° to V.P. Draw the projections when top end of the prism is nearer to V.P.

12. **(S-23)** A cylinder with 50 mm base diameter and 80 mm long axis, is lying on a generator on the H.P. with its axis parallel to the V.P. It is cut by an A.I.P. inclined at 30° to the H.P. and passes through a point on the axis 30 mm from one of its ends. Draw its anyone sectional top view.

13. **(W-23)** Hexagonal pyramid of side of base 25 mm and height 55mm is resting on HP on its base. It is cut by a plane parallel to HP and perpendicular to VP and cutting the axis at 25 mm above the base. Draw its projection.

Module 7: Development of Surfaces (Drawing)

1. **(W-24, W-23)** A hexagonal prism, edge of base 25 mm and axis 65 mm long, rests with its base on H.P such that one of its rectangular faces is parallel to V.P. It is cut by a plane perpendicular to V.P and parallel to HP and 30mm above HP. Draw the development of the lateral surface of the truncated prism.

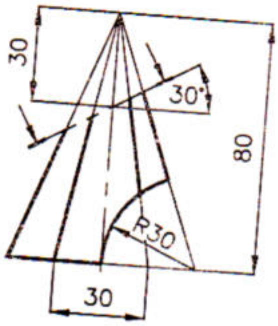
2. **(W-24)** A square pyramid, base 40 mm side and axis 65 mm long, has its base on the HP and all the edges of the base equally inclined to the VP. It is cut by a section plane, perpendicular to the VP, inclined at 45° to the HP and bisecting the axis. Draw its development.

3. **(S-25)** A cylinder of 40 mm diameter and 60 mm height is resting on its base on HP. It is cut by a section plane inclined at 45° to HP and passing through the top end of the axis. Draw the development of the lateral surface of the truncated cylinder.

4. **(W-18)** Draw the development of pentagonal prism of side 30mm and height 60mm, when one of the edges of the base is perpendicular to VP.

5. **(S-19)** Fig.2 shows the front view of a cut hexagonal pyramid. Draw the development of the lateral surface of the remaining portion of the pyramid.

   [Refer to Fig. 2 from Paper: Summer 2019]



6. **(W-19)** A right regular pentagon prism, edge of base 30 mm and height 75 mm resting on its base on HP, is cut by a section plane inclined to HP at 45^c and meeting the axis at a distance of 18 mm from its top end. Develop the outside surface of the cut prism.

7. **(S-20)** Draw the development of the lateral surfaces of a square pyramid, side of base 25 mm and height 50 mm, resting with its base on HP and an edge of the base is parallel to VP.

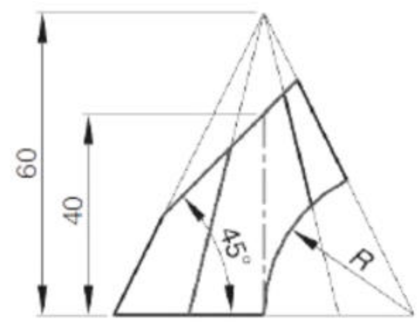
8. **(W-21)** Draw the development of lateral surfaces of a pentagonal prism with edge of base 40 mm and height 90 mm, kept on H.P. on its edge of base with an angle of base parallel to V.P.

9. **(W-22)** Develop the complete surface of a square prism of side of base 40 mm and height 80 mm.

10. **(W-22)** Draw the development of the lateral surface of a square pyramid, side of base 30 mm and height 50 mm, resting with its base on H.P. All edge of the base are equally inclined to V.P.

11. **(S-23)** A hexagonal pyramid, 30 mm base side and 60 mm long axis rests on the H.P. with a side of base parallel to V.P. It is cut by planes perpendicular to V.P., to obtain the front view as shown in Fig. 1. Draw the development of the lateral surface of the retained solid.

    [Refer to Fig. 1 from Paper: Summer 2023]

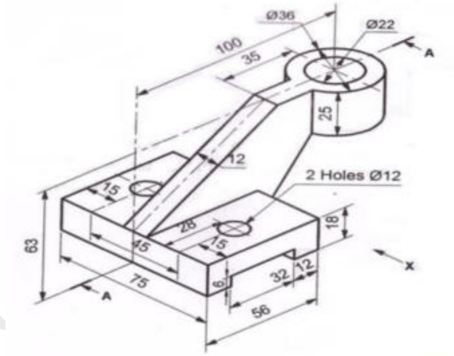


12. **(S-24)** Draw Development of lateral surfaces of following solids. (1) Cylinder (2) Cone (3) Pentagonal Prism (4) Cube

Module 8: Orthographic Projections (Drawing)

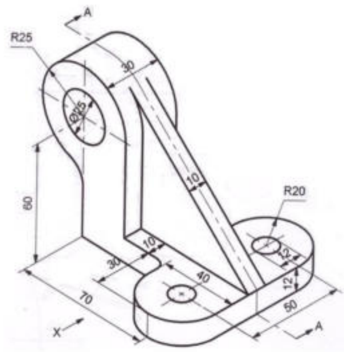
1. **(W-24)** Fig-1 shows pictorial view of an object with using particular using any one dimensioning method. Draw following views by first angle projection method (a) Sectional elevation at AA (b) Top view (c) Right hand side view.

   [Refer to Fig.-1 from Paper: Q.5(b) Diagram, Winter 2024]



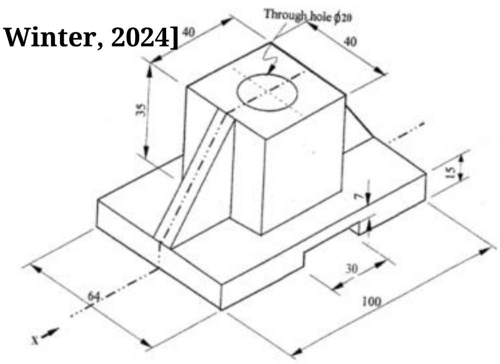
2. **(W-24)** Fig-2 shows pictorial view of an object with using particular any one dimensioning method. Draw following views by first angle projection method (a) Sectional elevation at AA (b) Top view (c) Left hand side view.

   [Refer to Fig.-2 from Paper: Q.5(b) OR Diagram, Winter 2024]



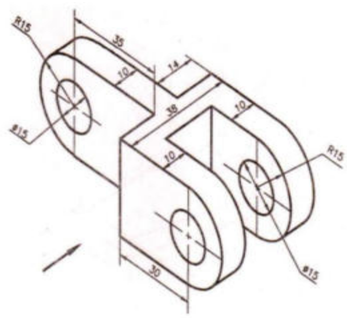
3. **(W-24)** Draw Front view, Top View and Right hand side view using 1^{st} angle projection method.

   [Refer to Unlabeled Figure from Paper: Q.5(b) Diagram, Winter, 2024]



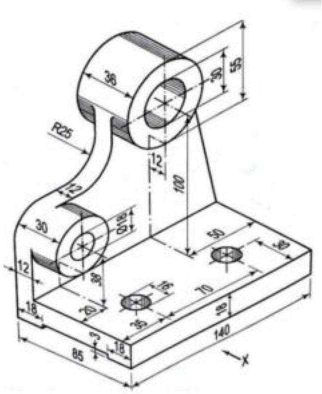
4. **(S-25)** Fig. 1 shows pictorial view of an object. Draw Sectional Elevation from X and Plan. In the figure 1, the arrow shows "X" direction.

   [Refer to Figure 1 from Paper: Summer 2025]



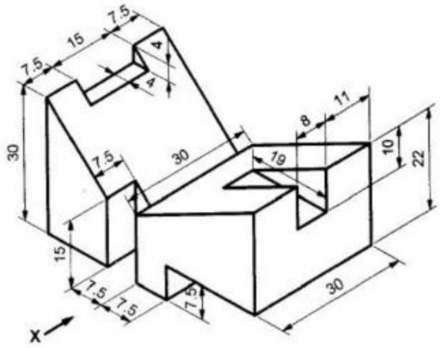
5. **(S-25)** Draw the front view looking from X-direction and plan of a object shown in Figure 2 using first angle projection method.

   [Refer to Figure 2 from Paper: Summer 2025]



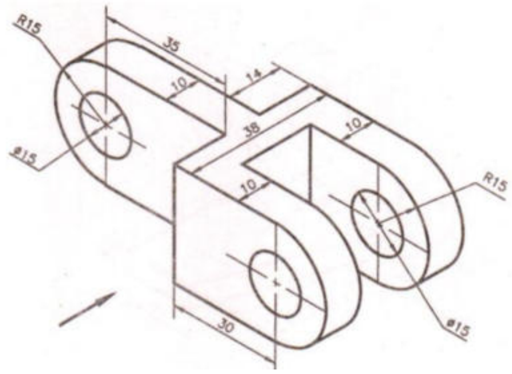
6. **(W-18)** Draw the (i) Front view (ii) Right hand side view and (iii) Top view of Fig. 02 in first angle projection method. Consider length as 50mm in direction of X.

   [Refer to Fig. 02 from Paper: Winter 2018]



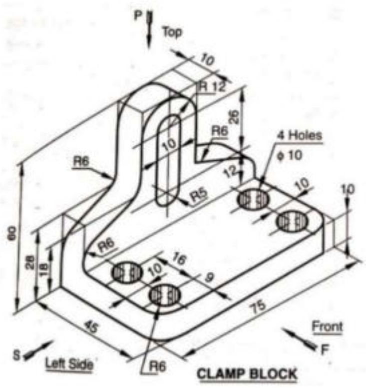
7. **(S-19)** Using third angle projection method draw right hand side view for the object shown in Fig. 3. Using third angle projection method draw front view and top view for the object shown in Fig. 3.

   [Refer to Fig. 3 from Paper: Summer 2019]



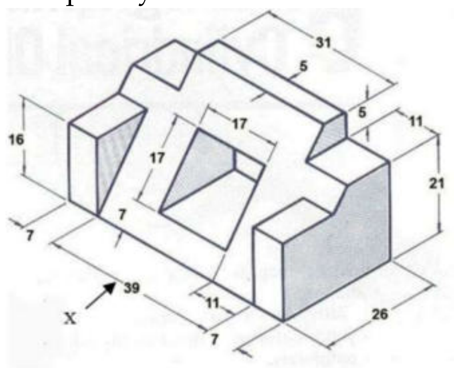
8. **(W-19)** Draw and dimension completely, the following views according to first angle projection method and to full size scale of a given figure 2. Front view looking in the direction of arrow F, Side view looking direction of arrow S, Top view looking the direction of arrow P.

   [Refer to Figure 2 from Paper: Winter 2019]



9. **(W-19)** Draw the front view looking in direction of arrow according to first angle projection method to full size scale of a given figure 3 with completely dimension.

   [Refer to Figure 3 from Paper: Winter 2019]



10. **(S-20)** Draw front view, top view and right-hand side view of the object given in Figure-2 using first angle projection.

    [Refer to Figure-2 from Paper: Summer 2020]

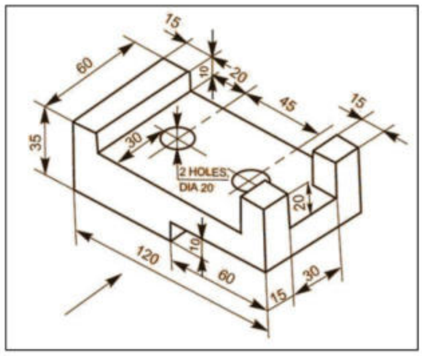
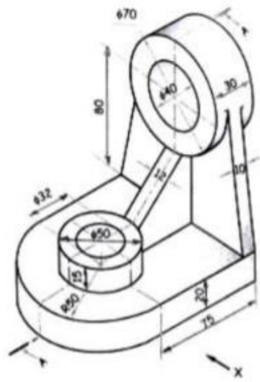


Figure-2 [Q4 (c)]

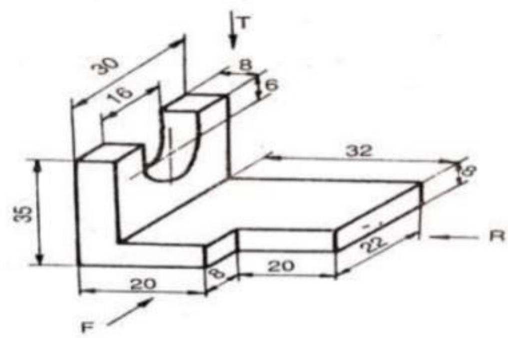
11. **(S-20)** Figure-3 shows the pictorial view of an object, draw the following views using first angle method of projection. Also give the important dimensions. (i) Sectional front view, along section A-A (ii) Top View

    [Refer to Figure-3 from Paper: Summer 2020]



12. **(W-21)** Figure No. 2 shows pictorial view of an object. Draw Front view looking in the direction F and also draw any one side view.

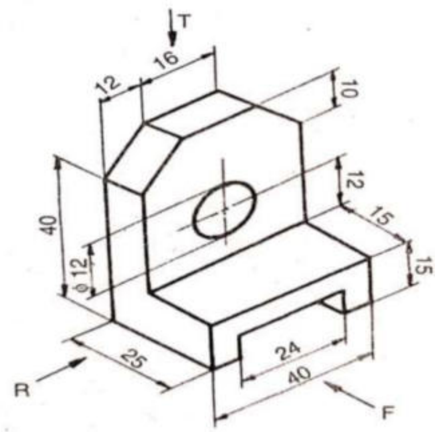
    [Refer to Figure No. 2 from Paper: Winter 2021]



13. **(W-21)** Figure No. 3 shows pictorial view of an object. Draw Front view looking in the

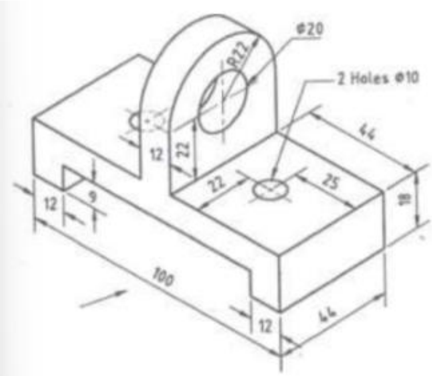
direction F and also draw any one side view.

[Refer to Figure No. 3 from Paper: Winter 2021]



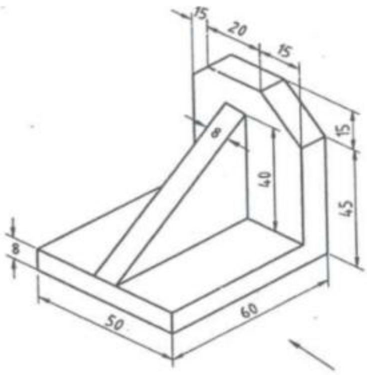
14. **(W-22)** Draw front view of an object shown in figure 2 using first angle projection methods. Draw top view and left hand side view of object shown in figure 2 using first angle projection methods.

[Refer to Figure 2 from Paper: Winter 2022]



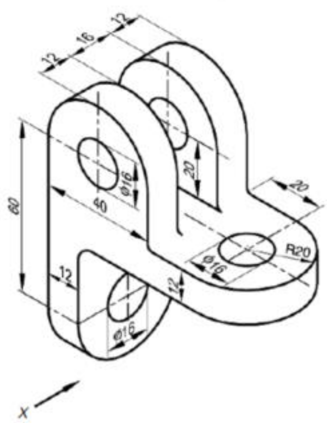
15. **(W-22)** Draw front view of an object shown in figure 3 using first angle projection methods. Draw top view and left hand side view of object shown in figure 3 using first angle projection methods.

[Refer to Figure 3 from Paper: Winter 2022]



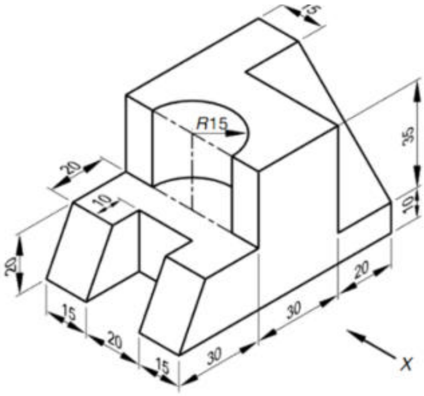
16. **(S-23)** Draw any three views using orthographic projections of fig 2.

[Refer to Fig. 2 from Paper: Summer 2023]



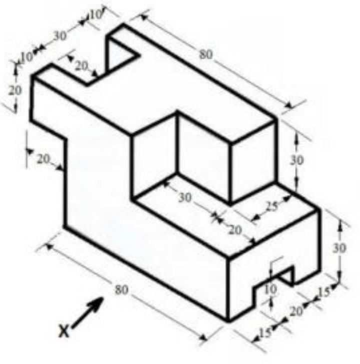
17. **(S-23)** Draw the Front view of Fig. 4 looking in the X direction and also draw any one side view.

[Refer to Fig. 4 from Paper: Summer 2023]



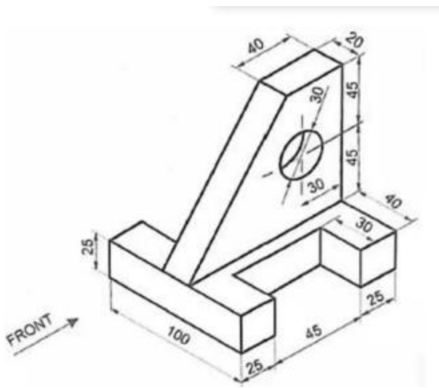
18. **(W-23)** For the figure shown below, draw plan, elevation and right hand side view using third angle projection method.

    [Refer to Unlabeled Figure from Paper: Q.5(b) Diagram, Winter 2023]



19. **(S-24)** Draw orthographic projection for the given figure.

    [Refer to Unlabeled Figure from Paper: Q.5(b) Diagram, Summer 2024]

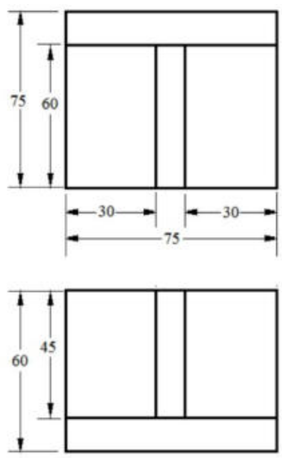


Module 9: Isometric Projections (Drawing)

1. **(W-24, S-24)** Draw isometric scale for the line of length 100mm. Mark 53mm on it.

2. **(W-24)** Draw isometric drawing of the following views.

   [Refer to Views from Paper: Q.5(c) Diagram (Page 4 of 4), Winter 2024



3. **(W-18)** Draw isometric view of the Fig.03 given below.

[Refer to Fig.03 from Paper: Winter 2018]

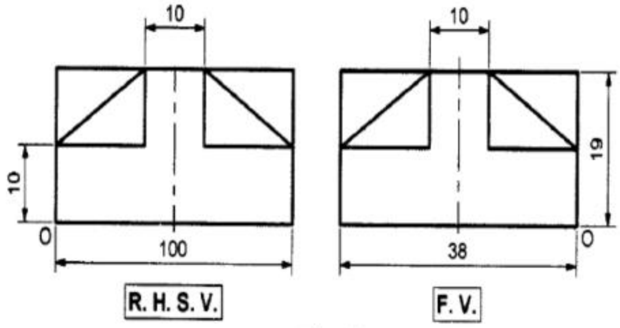


Fig.03

4. **(S-19)** Draw isometric view from the orthographic projection shown in Fig. 4.

   [Refer to Fig. 4 from Paper: Summer 2019]

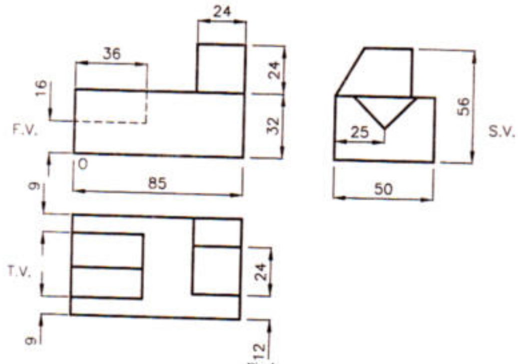


Fig.4

5. **(W-19)** Draw the isometric scale of 80 mm long line and show 66 mm isometric length on it.

6. **(W-19)** Draw the isometric view of the cone of 48 mm base diameter and 56 mm axis height.

7. **(W-19)** Figure 4 shows three orthographic views according to first angle projection method of an object. Draw its isometric projection.

   [Refer to Figure 4 from Paper: Winter 2019]

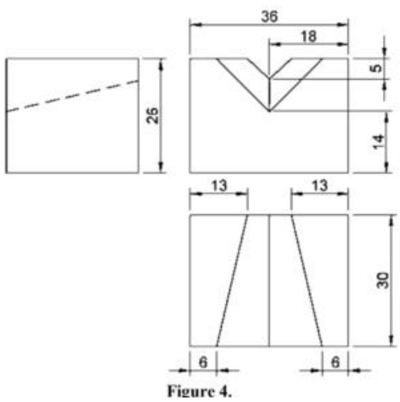


Figure 4.

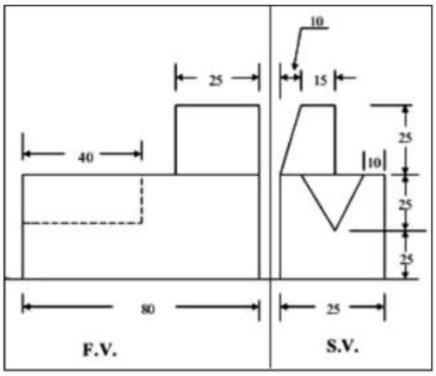
8. **(S-20)** Draw isometric circle on the three side of cube of 60 mm dimension.

9. **(S-20)** Draw isometric drawing of an object whose projections are given in Figure-4.

   [Refer to Figure-4 from Paper: Summer 2020]

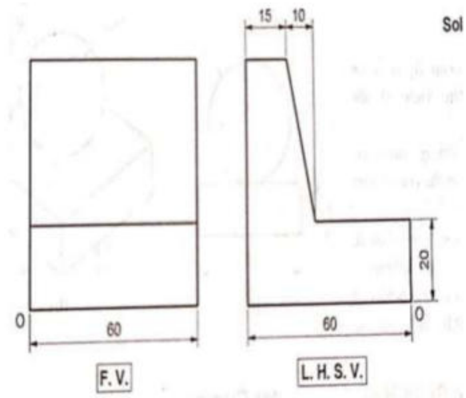
10. **(S-20, W-22)** Draw an isometric scale of 100 mm length and show 30 and 60 mm length on the scale.
11. **(S-20)** Draw isometric drawing of an object whose projections are given in Figure-5.

    [Refer to Figure-5 from Paper: Summer 2020]



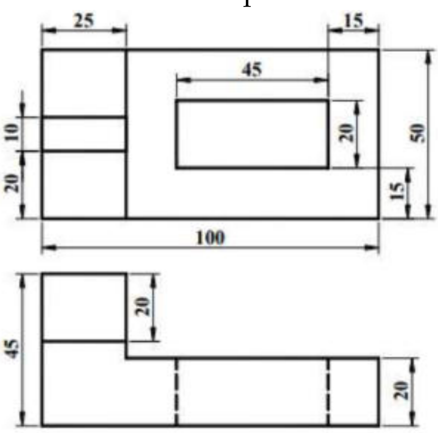
12. **(W-21)** Draw the pictorial view of truncated hexagonal pyramid.
13. **(W-21)** Figure No. 4 shows front view and top view of an object. Draw isometric view. Total height cover by the object is 60 mm.

    [Refer to Figure No. 4 from Paper: Winter 2021]



14. **(W-22)** Draw isometric scale showing 100 mm isometric length equivalent to 100mm normal length.
15. **(W-22)** Draw isometric projection of an object whose front view and top view are shown in figure 4.

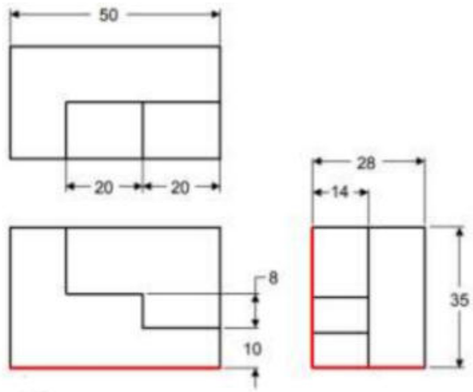
    [Refer to Figure 4 from Paper: Winter 2022]



16. **(W-22)** Draw isometric shape of a circle having 60 mm in diameter using four centre methods.

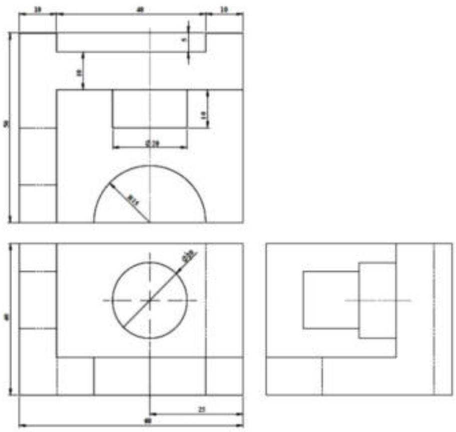
17. **(W-22)** Draw isometric projection of an object whose front view, top view and side view are shown in figure 5.

    [Refer to Figure 5 from Paper: Winter 2022]



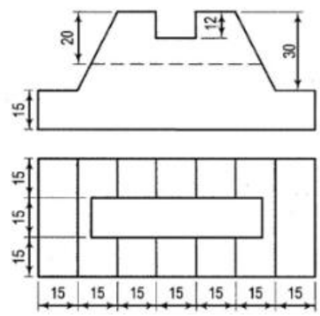
18. **(S-23)** Draw isometric view of given fig. 3.

    [Refer to Fig. 3 from Paper: Summer 2023]



19. **(W-23)** Draw isometric view of the following figure.

    [Refer to Unlabeled Figure from Paper: Q.5(b) OR Diagram, Winter 2023]



20. **(S-24)** Draw Isometric view of Circle of 50mm diameter, Pentagon of 30 mm side, a cube of 50mm side and a hexagon of 25 mm side.

21. **(S-24)** Draw isometric view for given figure.

[Refer to Unlabeled Figure from Paper: Q.5(c) OR Diagram, Summer 2024]